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Au nanoparticle-encapsulated hydrogel substrates for robust and reproducible SERS measurement

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A reliable and reproducible surface-enhanced Raman scattering (SERS) measurement utilizing Au nanoparticle-encapsulated hydrogels as a substrate has been demonstrated. A hydrogel matrix was adopted to: (i) take advantage of its excellent water uptake capacity for facile access of an analyte into the substrate and (ii) securely hold Au nanoparticles. Silica-coated Au (Au@SiO_2) nanoparticles were initially prepared and uniquely used as an initiator as well as a cross-linker for the polymerization of acrylic acid to synthesize Au nanoparticle-encapsulated hydrogels. Then, the outer silica layer of the Au nanoparticles in the hydrogel was etched out using hydrofluoric acid (HF) to make it possible for an analyte to approach the surface of the Au nanoparticles for generation of the SERS signal. In parallel, locally occurring SERS signals over the hydrogel were integrated using a wide area illumination scheme capable of covering a large area to improve quantitative representation of analyte concentration. To evaluate reproducibility of the proposed method, 6 independent hydrogels were prepared every two months over one year and then Raman spectra of 2-naphthalenethiol (2-NAT) captured hydrogels were collected. The resulting SERS intensities of 2-NAT acquired at each concentration were reproducible and clearly increased according to the elevation of 2-NAT concentration.

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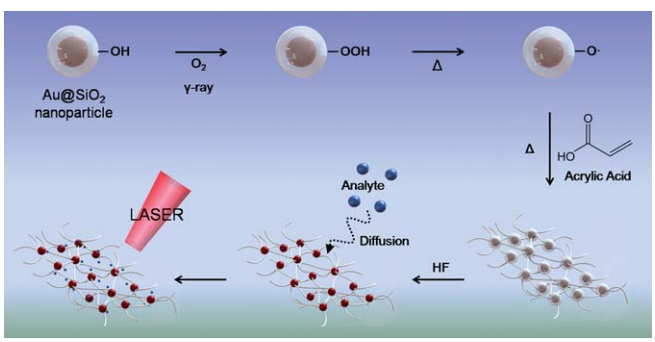
Introduction

Surface-enhanced Raman scattering (SERS) has been recently utilized as a powerful analytical tool due to its ultra-high sensitivity.[1–6] Widely different types of nanostructures made of diverse metals such as Au and Ag have been extensively investigated for SERS measurements.[7–9] Recently, the strategy of directly embedding nanostructures in a polymer matrix for SERS substrates has drawn attention since it could be one of the practical approaches to make SERS measurements more reliable for many analytical applications.[10–15] Several SERS substrates, such as Ag nanoparticle-loaded agarose gel, Ag nanoparticle- or Ag nanorod-embedded PVA nanofibers, and a capillary tube filled with a Ag nanoparticle-embedded polymer monolith, have recently been demonstrated.[10–15]

The requirements of a reliable nanostructure-holding polymer matrix for SERS measurement are facile access of an analyte into a matrix as well as physical sustainability of the polymeric structure without serious degradation or rupture upon sample uptake. A hydrogel is a suitable choice for the polymer matrix since it has a superior ability of water uptake;[16,17] therefore, it has been effectively used in Raman spectroscopy as a substrate for surface enhanced Raman optical activity (SEROA) to study enantiomers (D- and L-ribose)[18] and nondestructive identification of colorants in art works by SERS.[19]

In this publication, a unique Au nanoparticle-encapsulated composite hydrogel has been demonstrated as a reliable SERS substrate. The synthesis of a hydrogel was basically based on a previous method that utilized silica particles as an initiator as well as a cross-linker.[20] The graphical description for the preparation of the Au nanoparticle-encapsulated hydrogel and SERS measurement is provided in Scheme 1. For the synthesis of the Au nanoparticle-encapsulated poly(acrylic acid) (PAA) hydrogel,



Scheme 1 Graphical description of the overall procedure for the preparation of Au nanoparticle-encapsulated hydrogels.

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we used silica coated gold (Au@SiO_2) nanoparticles instead of just silica particles. By this way, Au nanoparticles could be effectively incorporated into the hydrogel without complicated steps. Then, the outer silica layer was etched out using hydrofluoric acid (HF) solution to make an analyte accessible to the surface of the Au nanoparticles for SERS measurement. The detail procedure is described in the Experimental section. For optimal analytical performance, the silica-etching time and concentration of Au nanoparticles in the hydrogels were properly determined.

SERS is a highly localized phenomenon, so acquired Raman intensity may vary greatly depending on where an analyte is positioned on the SERS-generating nanostructure. Since the size, shape and distribution of nanostructures in a substrate are variable, individual SERS intensities arising from analytes differ substantially.[21] In this situation, where only a tiny area is sampled for SERS acquisition, the resulting intensity will fluctuate depending on where the laser illuminates on the substrate. This is a crucial factor that affects the reproducibility of SERS-based quantitative analysis. The acquisition of SERS spectra by covering a large area on a substrate could effectively alleviate the problem of intensity fluctuation leading to degraded reproducibility of SERS measurement. A wide area illumination scheme capable of covering a large sampling area of 28.3 mm^2 (illumination diameter: 6 mm) was employed in this study,[22] and the size of the Au nanoparticle-encapsulated hydrogel was complied to fully accommodate the illumination area. Using 6 Au nanoparticle-encapsulated hydrogels separately prepared over a one-year period as described above, different concentrations of 2-naphthalenethiol (2-NAT) were measured and the acquired Raman intensities of the 2-NAT band were investigated to assess the long-term reproducibility of the proposed analytical method. To demonstrate analytical utility of the hydrogel, it was used to detect paraquat, a typical herbicide.

Experimental

Synthesis of Au nanoparticles

All reagents used in the study were of the highest available grade, purchased from Sigma-Aldrich, and used as received. The citrate reduction method was used to synthesize Au nanoparticles.[23] Fig. 1(a) shows a scanning electron microscope (SEM) image of the synthesized Au nanoparticles and the corresponding size distribution. The average size was 16.24 ± 2.09 nm with maximum extinction at 520 nm, as shown in the black spectrum in Fig. 1(c). A visible spectrum was collected using a UV/VIS spectrometer (Agilent Technologies), equipped with a photodiode array (PDA) (spectral resolution: 2 nm). SEM and transmission electron microscope (TEM) images were obtained using a Hitachi S-4800 and a JEOL JEM-2100F, respectively.

Synthesis of Au@SiO_2 nanoparticles

Au@SiO_2 nanoparticles were synthesized based on a previously described method.[24] Initially, 40.0 mL of a Au nanoparticle solution was diluted by mixing with 40.0 mL of water.

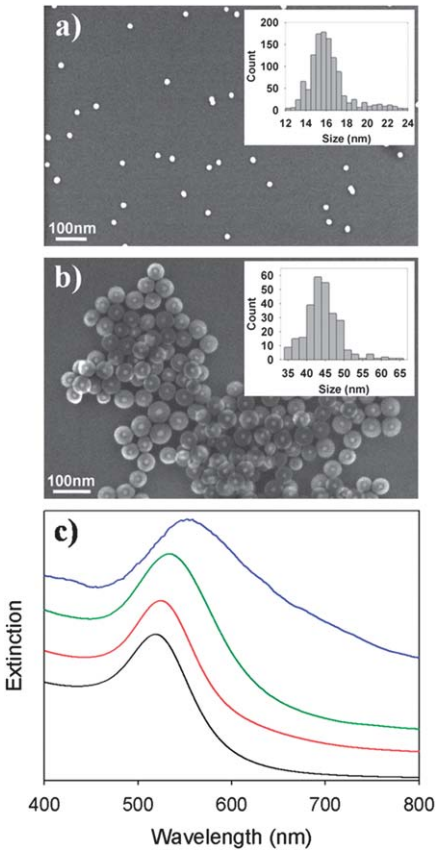

Fig. 1 SEM images of (a) Au nanoparticles and (b) Au@SiO_2 nanoparticles. The corresponding size distribution is shown in each image. Visible extinction spectra (c) corresponding to Au nanoparticles (black), Au@SiO_2 nanoparticles (red), Au@SiO_2-encapsulated hydrogel (green) and Au nanoparticle-encapsulated hydrogel (blue). The spectra are offset for clarity.

Then, Amberlite resin was added to the solution to reduce the ionic strength to a desired level. After the removal of the Amberlite resin, the pH of the solution was adjusted to 5.0. To make the nanoparticles vitreophilic, 0.5 mL of 1 mM 3-aminopropyl silane (APS) was added. After allowing APS to react with Au nanoparticles over 15 minutes, 40 μL of a 27.0 wt% silicate solution was added and stirred for 24 hours. To make the silica layer denser, 99.9% ethanol was added into the solution followed by stirring at room temperature over 24 hours. Finally, Au@SiO_2 nanoparticles were separated by centrifugation and re-dispersed in water for further use. Fig. 1(b) shows a SEM image of the Au@SiO_2 nanoparticles and the corresponding size distribution. The average size was 44.35 ± 4.77 nm. As shown, a silica layer is clearly formed around the Au nanoparticles. After the formation of the silica layer, the band slightly shifts to 525 nm (red color), as shown in Fig. 1(c).

Synthesis of Au nanoparticle-encapsulated hydrogel

For the synthesis of the Au nanoparticle-encapsulated PAA hydrogel, a previous method utilizing macromolecular microspheres was adopted.[16,20] The choice of PAA-based hydrogel in

this study was based on several merits. It is more viscoelastic compared to other hydrogels, so its structure is more sustainable under the repetition of swelling–de-swelling in an actual analysis. Also, as demonstrated in several previous publications, it has a small Raman cross-section, thereby minimizing interference from background signals.[18,25,26] As shown in Scheme 1, a solution of Au@SiO_2 nanoparticles was initially irradiated by Co^{60} γ-rays (dose rate: 10 kGy h^{-1}) over 2 hours in ambient conditions to generate peroxide (–OOH) on the outer surface of the silica layer. Immediately after irradiation, nitrogen gas was bubbled into the solution to remove dissolved oxygen to inhibit unwanted reactions between peroxide and oxygen. Then, the solution was heated to 40 °C to convert peroxides to oxygen radicals. The oxygen radicals initiated the homo-polymerization of acrylic acids to form Au@SiO_2-encapsulated hydrogels as shown in Fig. 2(a). The size of the Au nanoparticle-encapsulated hydrogel was 8.3 mm in diameter and 4.0 mm in thickness. The note "22 nM" (Au nanoparticle concentration) printed on a white paper was placed behind the hydrogel to illustrate the transparency. For access of an analyte to the surface of Au nanoparticles, the silica layer in the Au@SiO_2-encapsulated hydrogel was etched out using a 5 vol% HF solution. The silica-etched hydrogel is shown in Fig. 2(b). The extinction spectra acquired from the Au@SiO_2-encapsulated hydrogel before (green) and after silica-etching (blue) are shown in Fig. 1(c).

SERS measurement

For the collection of SERS spectra, initially solutions with 5 different concentrations of 2-NAT (50 nM, 500 nM, 5 μM, 50 μM and 500 μM) and 3 different concentrations of paraquat (0.1 μM, 10 μM and 1 mM) were prepared. Then, Au-encapsulated hydrogels were immersed in these solutions over 10 minutes. The swelled hydrogels were taken out and allowed to dry at ambient conditions over 30 minutes. The volume of swelled hydrogels returned nearly to the initial conditions. SERS spectra of these analyte-captured hydrogels were collected.

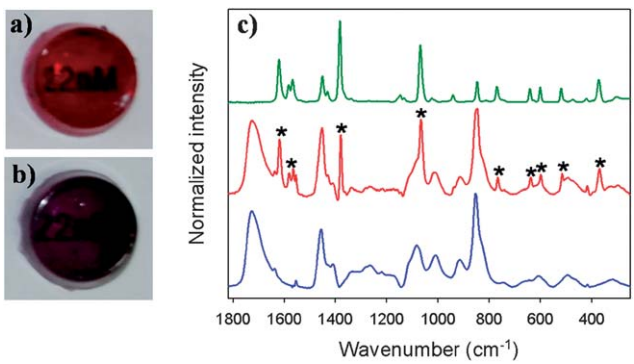

Fig. 2 (a) Pictures of the Au@SiO_2-encapsulated hydrogel and (b) the corresponding silica-etched hydrogel. The concentration of Au nanoparticles in the hydrogel was 22 nM. (c) Raman spectra corresponding to the Au@SiO_2-encapsulated hydrogel (blue), 50 μM 2-NAT-captured silica-etched hydrogel (red) and 50 μM 2-NAT acquired using only 22 nM Au nanoparticle solution (green). The asterisks on the red spectrum indicate the 2-NAT peaks.

To investigate the generation of the SERS signal from the analyte-captured Au nanoparticle-encapsulated hydrogel on a microscopic scale, a Raman microscope (Kaiser Optical System, Inc., Ann Arbor, MI, USA) equipped with a 785 nm laser was used. For quantitative analysis, a wide area illumination scheme capable of illuminating laser radiation (785 nm) into a large area (28.3 mm^2, diameter: 6 mm) was employed. A detailed description of the wide area illumination scheme can be found in our previous publication.[22] SERS spectra of 2-NAT-captured hydrogels were collected with an exposure time of 10 seconds for 10 scans (resolution: 4 cm^{-1}). The laser power at the sample was 200 mW, corresponding to a laser flux of 0.007 μW $μm^{-2}$.

Results and discussion

Optimization of silica-etching time

As pointed out in the Experimental section, the silica layer of Au@SiO_2 nanoparticles in the hydrogel physically blocks the access of an analyte (2-NAT in this study) to the Au nanoparticle surface, it should be properly removed without degrading the polymeric structure. A simple way to confirm silica-etching is the identification of 2-NAT peaks after immersion of the silica-etched hydrogel in a 2-NAT solution as shown in Fig. 2(c). The blue spectrum corresponds to the Au@SiO_2-encapsulated hydrogel before silica-etching. The spectral features are mostly from the vibrations of the polymeric hydrogel.[17,27] The middle red spectrum is acquired from the 2-NAT-captured hydrogel (2-NAT concentration: 50 μM). The 2-NAT peaks (marked by asterisks) are apparent on top of the hydrogel spectral feature, indicating the adsorption of 2-NAT on the Au surface by silica-etching. The features of 2-NAT peaks are in accordance with those found in the previous literature and the strongest 1380 cm^{-1} band (ring stretching) is solely used for further evaluation.[28] The SERS spectrum of 2-NAT acquired using only Au nanoparticles (top spectrum in green color) is also shown for comparison.

Since silica-etching initially occurs at the outer part of the hydrogel, it is necessary to achieve homogeneous silica-etching throughout the hydrogel for reproducible SERS measurement by optimally controlling the etching time. To identify the optimal conditions, the etching time in a 5% HF solution was varied from 5, 10, 30, 45 to 60 minutes. These silica-etched hydrogels were immersed in a 50 μM 2-NAT solution over 10 minutes and were dried in ambient conditions over 30 minutes. Then, the 2-NAT-captured silica-etched hydrogel was cut vertically along a diameter line using a razor to prepare a rectangular hydrogel slice (1.0 mm in thickness approximately). Raman spectra were collected using a microscope by horizontally crossing the middle of the sectioned rectangular hydrogel film with a point-to-point spacing approximately from 0.4 to 0.9 mm.

Nearly identical Raman intensities of 2-NAT across the line mapping indicates even silica-etching inside the hydrogel. However, if the intensities were inconsistent, *e.g.*, higher at both edges, it would be evidence of non-uniform silica-etching in the hydrogel. The peak height at the 1380 cm^{-1} band was used for evaluation since it was the strongest with minimal overlap with the bands of the hydrogel. Before calculation of the peak height,

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raw spectra were initially baseline-corrected at 8 different wavenumbers (1820, 1770, 1535, 1140, 700, 440, 320 and 250 cm^{-1}) and were normalized by dividing each baseline-corrected spectrum by the corresponding area.

Fig. 3 shows the variation of the peak height at 1380 cm^{-1} across the line mapping when the duration of silica-etching varies. The etching times are indicated by different colors. When silica-etching lasted less than 30 minutes, the intensities were stronger near both edges and no significant Raman signal was observed around the center. This clearly shows that silica-etching occurs predominantly at the outer part of the hydrogel when the etching time is insufficient. At an etching time of 45 minutes, the intensities around the center start to increase, while those at both edges remain higher. When the etching continues over 60 minutes, the intensities increase substantially and become similar across the mapping line, indicating uniform silica-etching. Consequently, one-hour silica-etching in a 5% HF solution was chosen as an optimum. At this condition, the color of the hydrogel changes from red to purple or deep red after silica-etching (see Fig. 2(a) and (b)). The color change after silica-etching is from the partial aggregation of Au nanoparticles by silica-etching responsible for the varied extinction spectral feature. Except for a minor variation in the softness, no noticeable physical variations were observed after silica-etching.

To alternatively identify the silica-etching, TEM images of the hydrogel before and after the etching were obtained, as shown in Fig. 4(a) and (b), respectively. After etching, the silica layers clearly disappeared and only the image of Au nanoparticles is observable. Both SERS line mapping across the sectioned hydrogel and TEM images ensure the etching of the silica layer on the Au nanoparticles in the hydrogel.

Optimization of Au nanoparticle concentration in hydrogel

The concentration of Au nanoparticles in the hydrogel is also important since it is one of the crucial factors governing the sensitivity as well as reproducibility of measurement. To find an optimal concentration of Au nanoparticles, 9 different hydrogels were prepared by varying the Au nanoparticle concentration from 6 nM (3.61×10^{15} particles per L) to 38 nM (2.29×10^{16} particles per L) with an increment of 4 nM (2.41×10^{15} particles per L). The concentration was estimated based on the extinction coefficient (ε) of 2.0×10^{8} M^{-1} cm^{-1} at 520 nm from the extinction spectrum of bare Au nanoparticles (see Fig. 1(c)), as similarly performed in a previous study.[29]

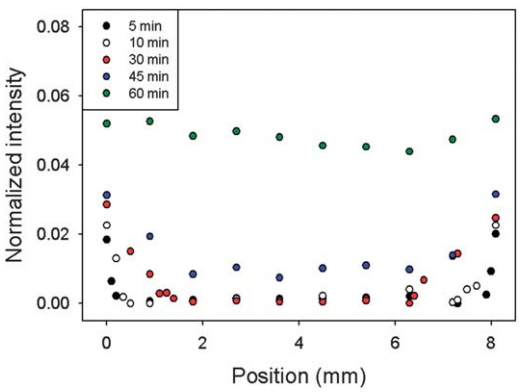

Fig. 3 Variation of the peak height at 1380 cm^{-1} across the line mapping as the duration of silica-etching varies. The etching times are indicated by different colors.

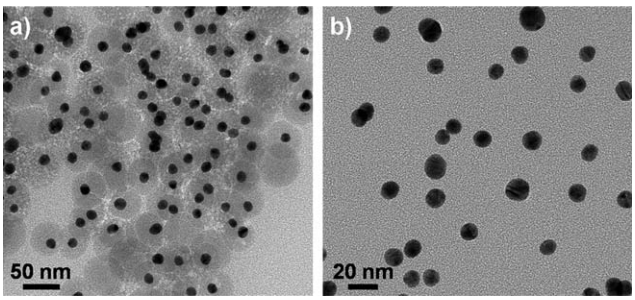

Fig. 4 TEM images of the Au@SiO_2-encapsulated hydrogel (a) before and (b) after etching of the silica layer. The concentration of Au particles in the hydrogel is 22 nM.

Pictures of Au nanoparticle-encapsulated hydrogels synthesized using different concentrations of Au nanoparticles are shown in Fig. 5. For simplicity, only 4 hydrogels containing 6, 14, 22 and 30 nM Au nanoparticles are shown. Similarly, these hydrogels were placed on the corresponding concentrations printed on white paper for evaluation of the transparency. The color changes with the increase of concentration, from light purple to dark purple. The extinction at 530 nm linearly increased according to the concentration of Au nanoparticles in the hydrogel (data not shown).

Higher concentrations of Au nanoparticles would be advantageous to enhance the sensitivity of the measurement, while the increased Au@SiO_2 concentration could vary the final physical properties of the hydrogel, and possibly influence the

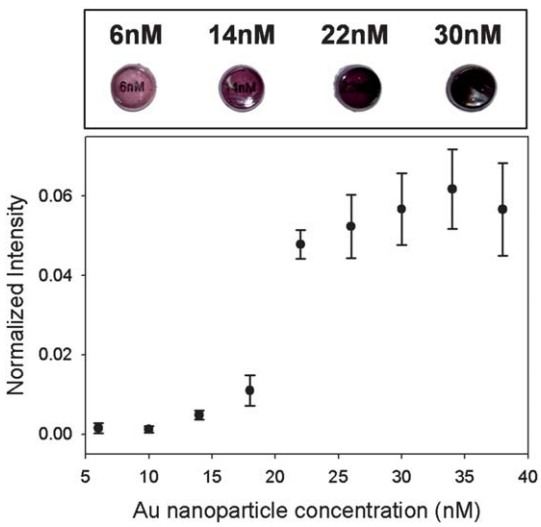

Fig. 5 Pictures of Au nanoparticle-encapsulated hydrogels synthesized with different concentrations of Au nanoparticles (top) and the intensity variation (bottom) of a 50 μM 2-NAT band at 1380 cm^{-1} when the concentration of Au nanoparticles changes from 6 to 38 nM. The error bars indicate the standard deviation of peak heights from 6 independent measurements using separately prepared hydrogels over one year.

analytical performance of hydrogel-based SERS measurement. These hydrogels were immersed in a 50 μM 2-NAT solution and the magnitude of peak intensities at 1380 cm^{-1} was examined. Fig. 5 shows the intensity variation of the 2-NAT band when the concentration of Au nanoparticles in the hydrogel changes from 6 to 38 nM. One Raman spectrum was acquired from each hydrogel using the wide area illumination (WAI) scheme by illuminating a laser at the center of the hydrogel. The error bars indicate the standard deviation of peak heights from 6 independent measurements at each concentration using separately prepared hydrogels over approximately one year (*i.e.*, the preparation of hydrogels and SERS measurements approximately every two months).

As shown, the intensity increases as more Au nanoparticles are present in the hydrogel. The intensity substantially increases at the concentration of 22 nM, while the increase of the peak intensity was rather marginal after this concentration. From the concentration of 26 nM, the hydrogel became too soft and then their physical texture among the separately prepared hydrogels started to vary greatly, resulting in inconsistent distribution of Au nanoparticles and/or possibly inducing aggregation. A noticeable observation is that the gel-to-gel reproducibility suddenly deteriorates above a concentration of 26 nM. The high concentration of Au@SiO_2 nanoparticles would not be desirable for synthesizing hydrogels with reproducible physical properties, while the polymeric property was nearly consistent for the hydrogels prepared with the concentrations below 22 nM. Consequently, the Au nanoparticle concentration of 22 nM providing both sensitivity and gel-to-gel reproducibility was selected for further analysis.

Measurement of different concentrations of 2-NAT

It was our expectation that Au@SiO_2 nanoparticles were fairly uniformly distributed throughout the hydrogel because they were involved in the polymerization of acrylic acid. For the confirmation, Raman mapping (9.0 × 9.0 mm^2) fully covering the 2-NAT-captured hydrogel (etching time: 60 minutes, Au nanoparticle concentration: 22 nM, 2-NAT concentration: 50 μM) was performed. The point-to-point mapping distances in x and y directions were 200 and 200 μm, respectively, thereby resulting in a total of 2025 mapping points (45 × 45 points). The intensity of the 1380 cm^{-1} band was again used to construct the mapping image.

Fig. 6 shows the corresponding Raman image of the 2-NAT-captured hydrogel. As indicated by green and orange colors, the variation of Raman intensities along the edge is slightly larger compared to those in the middle. This could be from a minor fluctuation in the Au nanoparticle population along the edge due to the surface tension between a solution containing Au@SiO_2 nanoparticles and the wall of the glass ampoule. Except for along the edge, the intensities are quite similar over the whole hydrogel. To quantitatively evaluate the intensity variation, the relative standard deviation (RSD) of intensities (total of 1368 spectra) covering the whole hydrogel was calculated. The resulting RSD was 15.3%, confirming fairly reproducible SERS intensities emanating from the hydrogel.

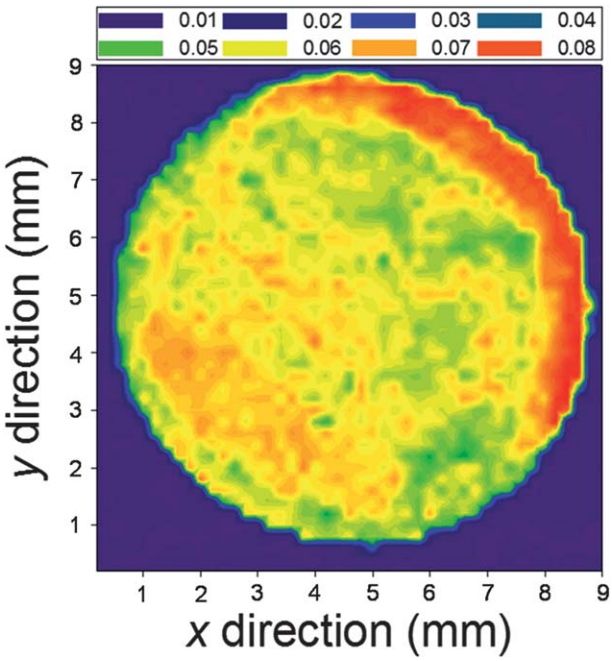

Fig. 6 Raman mapping image of a 50 μM 2-NAT-captured hydrogel constructed using the intensity of the 1380 cm^{-1} band. The point-to-point mapping distance in the x and y direction was 200 and 200 μm (total 2025 mapping points).

The reproducibility of the WAI-based spectral collection for a single hydrogel was also investigated. Approximately 51% of the total area of the hydrogel was covered for the measurement. A laser illuminated 5 different spots on the hydrogel (captured 50 μM 2-NAT), as described in Fig. 7(a). Raman spectra were collected for the opposite face by illuminating a laser in the same fashion. So, a total of 10 spectra were collected from the hydrogel. Fig. 7(b) shows the acquired intensities at the 1380 cm^{-1} band for each replicate measurement. Two colors indicate two opposite hydrogel faces exposed for spectral collection. No significant differences in the intensities among the replicate measurements were observed with a RSD of 2.5%. The large area coverage on the hydrogel for spectral collection leads to steady SERS intensities.

To evaluate quantitative performance, the Au nanoparticle-encapsulated hydrogels (concentration: 22 nM) were immersed in the solutions of 5 different 2-NAT concentrations (50 nM, 500 nM, 5 μM, 50 μM and 500 μM) over 10 minutes. After immersion, the hydrogels swelled approximately 236 ± 24% in volume. Although the degree of swelling of the PAA-based hydrogel is dependent on pH, there was only a minor difference in the degree of swelling among the hydrogels immersed in the sample solution of different concentrations in 10 minutes. The immersion period (10 minutes) was decided to fulfill three requirements: ensuring a clearly observable SERS signal, reproducible measurement insensitive to the degree of swelling and fast analysis. At this condition, the disk-shape was firmly maintained even after swelling without indication of rupture or significant deformation in the shape. Then, the volume of swelled hydrogels returned nearly to the initial condition after drying at ambient conditions over 30 minutes, confirming good structural sustainability of the hydrogel for analytical usage.

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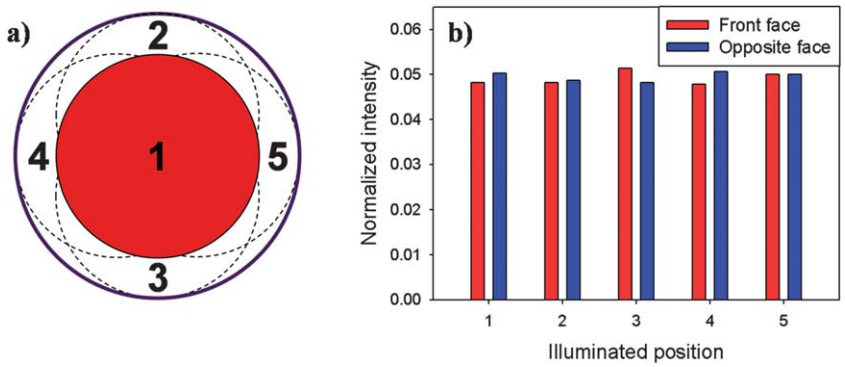

Fig. 7 (a) Description of 5 different illuminated spots on the hydrogel for spectral collection using the wide area coverage scheme and (b) the variation of intensities at a 1380 cm^{-1} band for 10 replicate measurements.

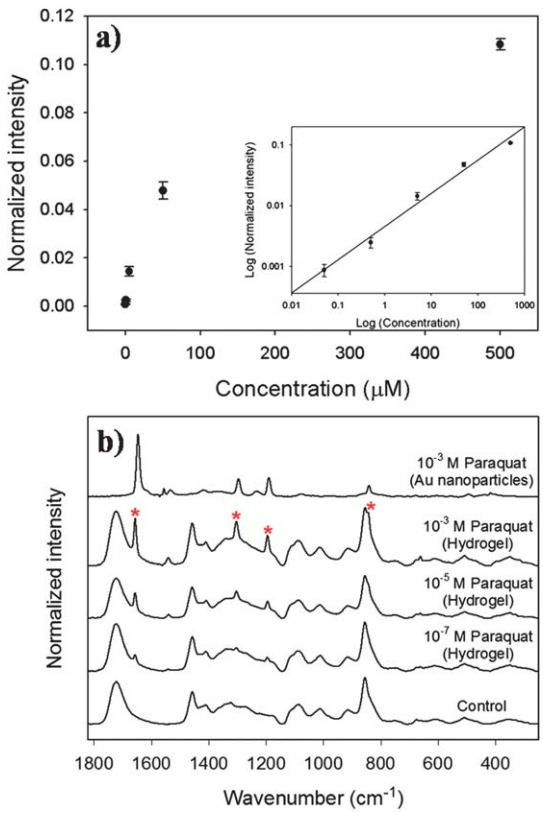

Fig. 8 (a) Response curve showing the variation of the peak intensity at 1380 cm^{-1} as a function of 2-NAT concentration. The inset shows the response when both axes are converted to a logarithm scale. The error bars were calculated based on 6 independent measurements. (b) SERS spectra corresponding to three different concentrations of paraquat (0.1 μM, 10 μM and 1 mM) as well as control. For comparison, the SERS spectrum of 1 mM paraquat acquired using Au nanoparticles is also shown (top spectrum). The asterisks indicate major peaks appeared on top of the hydrogel spectral feature.

At each concentration, 6 independent hydrogels prepared every two months over a one-year period were used for the measurement. The response curve showing the variation of the peak intensity at 1380 cm^{-1} as a function of 2-NAT concentration is shown in Fig. 8(a) (collection of one spectrum for each hydrogel). The intensity increases according to the concentration variation. When both axes are converted to a logarithm scale, the response is linear over the concentration range, as shown in the inset. The linear relationship between log(concentration) and log(intensity) was previously confirmed in a publication, when the adsorption of analyte molecules follows Langmuir isotherm.[30] At each concentration, the intensities are reproducible, as indicated by the small magnitude of the error bars. The achieved RSDs at 50 nM, 500 nM, 5 μM, 50 μM and 500 μM were 22.4, 19.9, 14.0, 7.5 and 2.2%, respectively. The average RSD was only 13.2%. The obtained gel-to-gel reproducibility is superior, although 6 independently prepared hydrogels were used for the evaluation.

To demonstrate practical applicability, we also used the hydrogels to detect paraquat, a typical herbicide. Fig. 8(b) shows SERS spectra corresponding to three different concentrations of paraquat (0.1 μM, 10 μM and 1 mM) as well as control. For comparison, the SERS spectrum of 1 mM paraquat acquired using Au nanoparticles is also shown (top). The asterisks indicate major peaks appeared on top of the hydrogel spectral feature. The peak intensities clearly increase with the elevation of paraquat concentration.

Conclusions

The combination of reliable Au nanoparticle-encapsulated hydrogels and spectral acquisition covering a large area on the hydrogel is effective for reproducible SERS measurements. The hydrogel allowed easy uptake of aqueous samples without serious deformation of its shape. SERS intensities obtained from 6 hydrogels prepared at separate periods over one year were reproducible, demonstrating superior analytical utility for further applications. The achieved results in this study could be one of the valuable advances in an effort to improve reproducibility of SERS measurements, while the sensitivity needs to improve further for more practical applications. Future research will explore the incorporation of other nanostructures, such as Au particles with larger size, nanorods or controlled aggregates with hotspots, in a hydrogel to eventually enhance the SERS signal.

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